

Data Intensive Assignment 3

June 2026

1 Requirements

Install the requirements with:

```
pip install -r requirements.txt
```

2 Run

Start MiniStack in one terminal:

```
ministack
```

In another terminal:

```
export AWS_ACCESS_KEY_ID=test
export AWS_SECRET_ACCESS_KEY=test
export AWS_DEFAULT_REGION=us-east-1

bash run.sh
```

3 Uploading a review for a quick test

For this, we provide a sample review under `src/sample_review.json`.

```
aws --endpoint-url=http://localhost:4566 \
s3 cp src/sample_review.json s3://review-app-raw/sample_review.json
```

Inspect the final analyzed review:

```
aws --endpoint-url=http://localhost:4566 s3 cp \
s3://review-app-analyzed/sample_review_analyzed.json -
```

Inspect customer counts and ban status:

```
aws --endpoint-url=http://localhost:4566 dynamodb scan \
--table-name review-app-users
```

An impolite review increments `impoliteReviewCount`. A customer is marked with `isBanned: true` after more than three impolite reviews.

4 Running dataset

For running the pipeline on the dataset to get the final results:

```
python3 src/scripts/run_dataset.py
```

5 Automated Integration Tests

We have a total of 5 tests - preprocessing, positive review, negative review, profanity, and banning of a user. Run all tests using:

```
python3 -m pytest src/tests -v
```